Economic Recovery After Wildfires: Evidence from County-Level Data in California

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Introduction

Wildfires have become more common and intense in California in recent years. They are known to destroy businesses and interrupt economic activity. However, the impacts on local economies are not fully understood. This project examines how local economies recover following major wildfire events. The research question is: How do employment levels in affected California counties differ compared to similar counties that did not experience major fires? To answer this question, we focus on the 2018 Camp Fire in Butte County and combine county-level wildfire data with employment data as economic indicators. For comparative analysis, we select near-by Shasta County, which did not experience a similar event during the same period, and analyze before and after changes and short- and medium-term economic impacts.

Figure 1. Highlighted study area of the research.

Background and Related Work

Wildfires are a known economic threat. Reports estimate California wildfires caused over $10 billion in insured losses in 2017 and 2018 each, including property and infrastructure losses and business shutdowns, as well as indirect ripple effects beyond local markets (Bay Area Council Economic Institute, 2023). A 2023 study investigating the extent of economic effects finds that wildfire damages affect regionally interlinked industries and service lines (Kim & Kwon, 2023). The UCLA Anderson Forecast reports on the scale of urban disruptions as Los

Angeles-area wildfires resulted in long-term business activity and labor force declines (Li & Yu, 2017-2025). Prior research demonstrates that local economic shocks can be effectively captured by tracking employment, as this indicator responds quickly to changes following major disruptions (Eckert et al., 2025). This supports our use of county-level employment as a valuable indicator for assessing wildfire impacts. Although the literature indicates significant, varied economic consequences, there is limited evidence on how wildfire impacts occur at the county level over multiple years, in particular in economically weaker, more rural counties. Insights into these developments is crucial for planning wildfire aid, recovery measures, and community vulnerability to long-term economic problems. Comparing Butte County with a similar control county, this project contributes to understanding the effects of major California wildfires.

Datasets

This study combined wildfire incident data with county-level employment data. Wildfire data were obtained from the California Department of Forestry and Fire Protection (CAL FIRE) incident dataset, which provides details on wildfire location, acres burned, incident dates, and geographic coordinates. We employed the CAL FIRE records from 2015 to 2020, identified major wildfire events, and focused on the 2018 Camp Fire in Butte County. To measure economic outcomes, we used employment data from the Quarterly Census of Employment and Wages (QCEW) by the U.S. Bureau of Labor Statistics. The QCEW dataset reports annual average employment at the county level across all industries. We utilized employment data for Butte and Shasta Counties 2015–2020 to compare pre- and post-fire labor market trends. Butte County was selected as the treatment group because it experienced the 2018 Camp Fire. Shasta County served as the control group, as it did not experience a major wildfire during the study period. The datasets were merged using county identifiers and year variables. The final dataset includes both annual and monthly employment levels for the treatment and control county.

Methods

We prepared the data, conducted exploratory data analysis (EDA), including trend and time-series analysis, and applied a Difference-in-Differences (DID) model. All datasets were cleaned, standardized, and integrated using Python as the primary data processing tool. Within the CAL FIRE incident dataset, the records were filtered to the years 2015–2020 and to variables identifying major wildfire events, including county, acres burned, and incident dates. Within the

QCEW employment dataset, both annual average and monthly employment for all industries at the county level were selected and reduced to the same time period. The Butte and Shasta Counties employment datasets were then combined and merged.

EDA was performed to gain an overview about data structure, capture trends, quantify evolution over time, and identify relevant patterns. State-wide wildfire incident counts by county and a geographic map were reviewed to contextualize the respective wildfire exposure of Butte and Shasta Counties. Visual comparisons of employment trends for both counties were used to assess overall trajectories and evaluate the parallel trends assumption. Because both counties share the same prevalent industries, employment trends were also examined for the two largest industries to determine whether industry-specific patterns aligned with aggregate trends.

A DID framework was selected as the inferential analytical model. DID is used to estimate the causal effect of a treatment, intervention, or event on periodic data. The method compares changes in outcomes over time between a treatment group affected by the intervention and a control group that was not affected.

The general DID regression equation is: $Y_{it} = \beta_0 + \beta_1 D_i + \beta_2 T_t + \beta_3 (D_i \times T_t) + \varepsilon_{it}$

Where

- Y_{it}: Outcome variable for unit i at time t
- D_i: Binary indicator for the treatment group
- T_t: Binary indicator for the post-treatment period
- $D_i \times T_t$: Interaction term
- β : The coefficients of each term

A key assumption of DID is the parallel trends assumption, which states that both groups (Butte and Shasta Counties) would have followed similar trends over time in the absence of treatment.

As the experimental setup, Butte County was designated as the treatment group based on the 2018 Camp Fire, a major wildfire with substantial economic impact. Shasta County was selected as control due to its similar regional characteristics, industry focus, and lack of major wildfire during the study period. Treatment variable is the wildfire event, outcome variable is annual average employment. Event year is 2018, pre-fire period is defined as 2015–2017, and post-fire period is 2019–2020. This setup facilitates isolating and comparing changes in employment before and after the wildfire and estimating the treatment effect using a DID framework.

Results: Exploratory Data Analysis

	County	Count
1	Riverside	320
2	Fresno	171
3	San Diego	165
3	Kern	165
5	San Luis Obispo	117
6	Los Angeles	113
7	San Bernardino	112
8	Siskiyou	104
9	**Butte**	**97**
10	**Shasta**	**92**

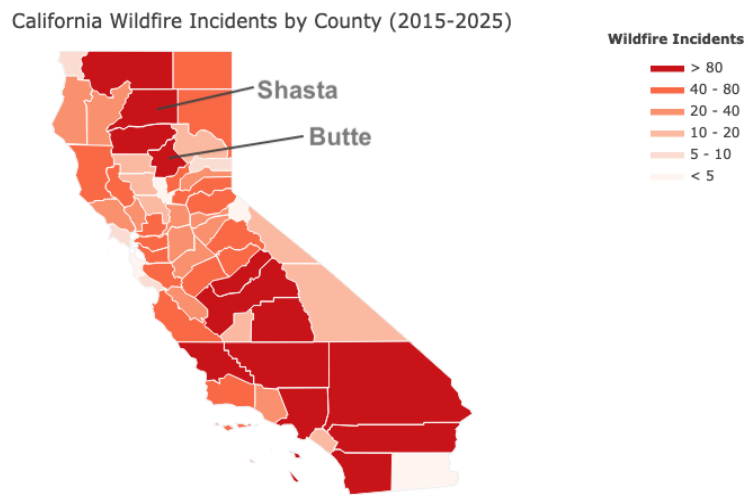


Figure 2. California counties with most wildfire incidents: top 10 based on CAL-FIRE records (left) and state-wide map showing counties by wildfire incidents between 2015 and 2025 (right).

Figure 2 presents the top 10 California counties by wildfire incidents from 2015 to 2020 and a statewide map highlighting geographic distributions of wildfire activity. Both Butte and Shasta rank in the top 10. The map depicts wildfire concentrations in few top northern and largely in very southern regions, and confirms both study counties' generally elevated wildfire occurrence.

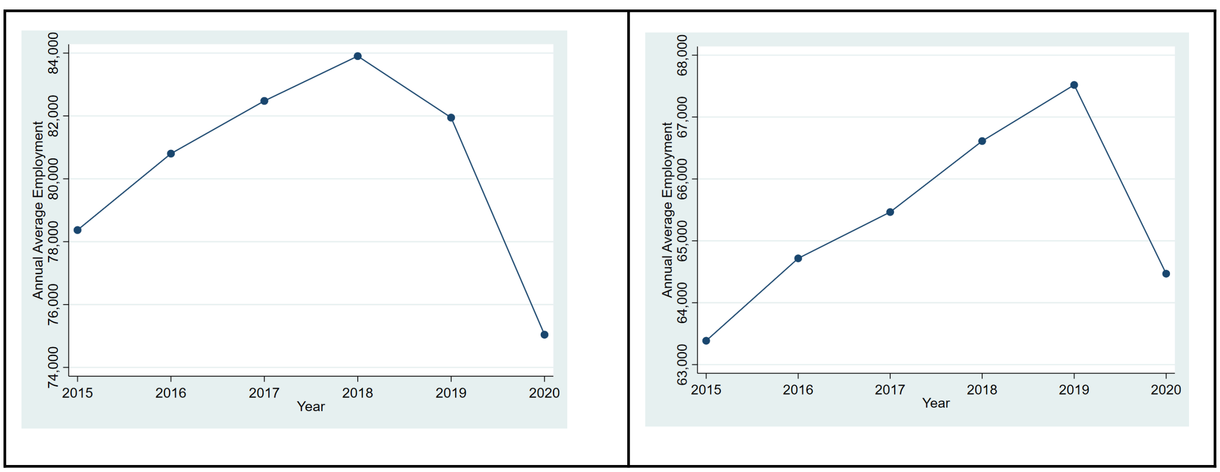


Figure 3. Employment trends 2015-2020 in Butte County (left) and Shasta County (right), showing the trajectory before and after the 2018 Camp Wildfire.

Figure 3 reveals annual employment trends for Butte and Shasta Counties from 2015 to 2020. Both counties experienced steady linear employment growth during the pre-fire period followed by a decline after the 2018 Camp Fire. Butte shows a steep decline in 2018 (time of Camp Fire) while Shasta's employment drops a year later. The sharp drop in 2019/2020 suggests prolonged economic disruption, potentially compounded by additional external shocks.

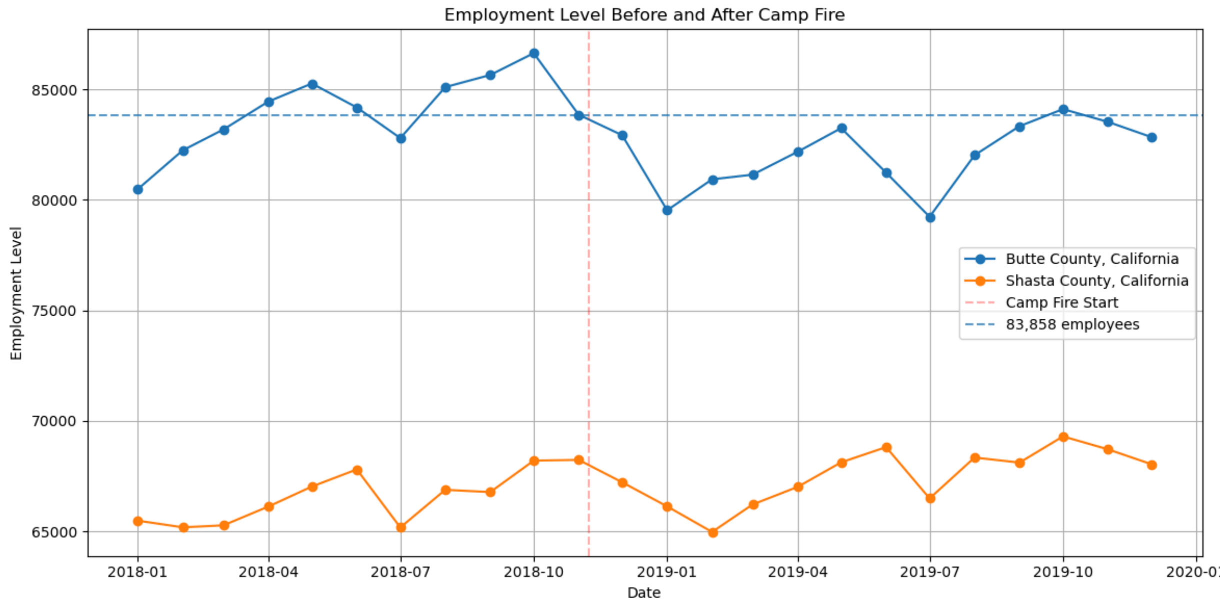
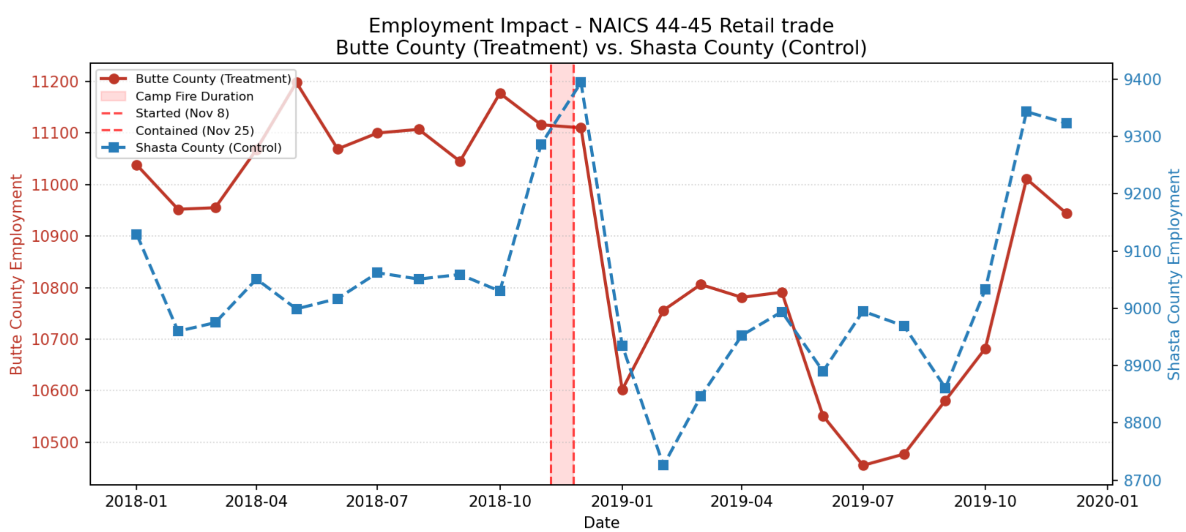


Figure 4. Employment Levels Butte vs. Shasta County before and after Camp Fire.

Figure 4 shows combined employment levels for Butte and Shasta Counties before and after the Camp Fire. The plot demonstrates the relative positions of both counties over time revealing changes around the event year. Butte County and Shasta County followed similar employment patterns from 2018 to 2019, supporting the parallel trends assumption required for DID analysis. The Camp Fire began on November 8, 2018, which is marked by the vertical red line.



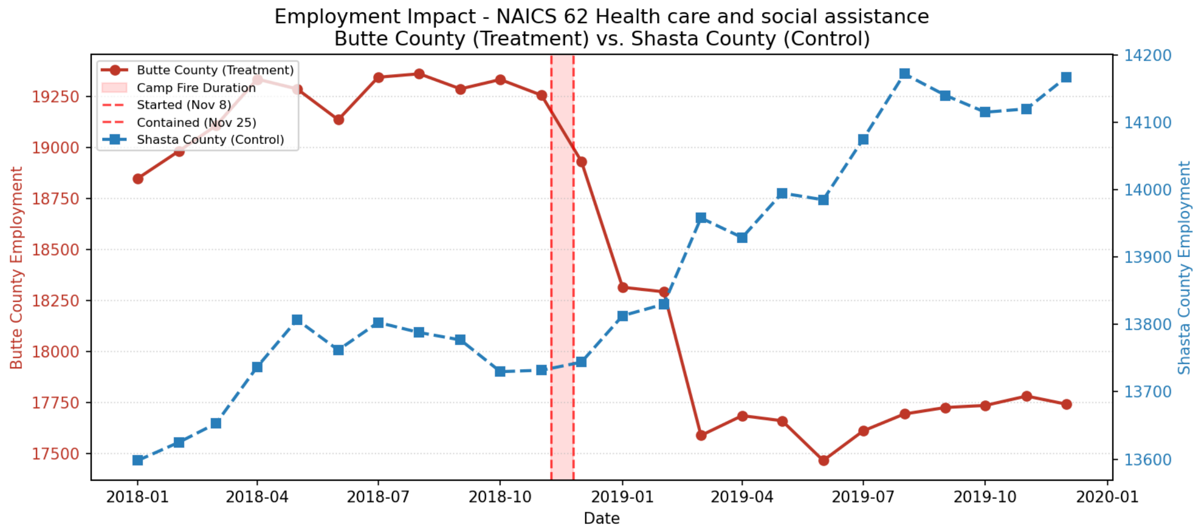


Figure 5. Tracking of employment impact of Camp Fire in treatment vs. control by top 2 industries.

Figure 5 presents employment trends for two dominant industries in both counties: healthcare and social assistance, and retail trade. These sectors were evaluated for their major employment impact. The plots elucidate how industry-specific employment developed over the study period and if sector-level patterns correspond with the aggregate trends shown in previous graphs.

Results: Difference-in-Differences Model

After running the DID analysis, the results suggest that the wildfire had a measurable negative effect on employment levels in Butte County. The estimated DID regression equation is:

$$Y_{it} = 66,390 + 17,610D_i + 1,159\,T_t - 3,010\left(D_1 \times T_t\right) + \varepsilon_{it}$$

	Coefficient	Std. Error	t-value	p-value
Intercept	66,390.00	458.42	144.823	0.000
Treated	17,610.00	648.304	27.17	0.000
Post	1,158.80	600.213	1.931	0.060
DID	-3,009.67	848.829	-3.546	0.001

The coefficients are interpreted as follows:

➔ The intercept coefficient (66,390) represents the baseline employment level in Shasta County before the Camp Fire.

➔ The treatment coefficient (17,610) indicates that, before the wildfire, Butte County had approximately 17,610 more employees on average than Shasta County.

➔ The DID interaction coefficient (−3,009.67) represents the estimated wildfire effect. This indicates that after the Camp Fire, Butte County experienced an additional decrease of approximately 3,010 employees relative to Shasta County after controlling for baseline county differences and common time trends.

The p-value for the DID coefficient is 0.001, indicating statistical significance (5% threshold). Therefore, the analysis suggests that the Camp Fire was associated with a substantial negative impact on employment levels in Butte County. The pre-fire employment trends observed in the EDA support the parallel trends assumption required for interpreting this estimate.

Discussion

The CAL FIRE incident dataset recorded 3,147 wildfires between 2015 and 2025. Figure 2 shows the ten counties with the most wildfire incidents over this period. Wildfire risk is shaped by several components, such as climate conditions, fuel characteristics, topography, and human activity, all of which affect ignition probability and fire spread. Although larger counties may be expected to have more wildfires, several smaller counties report relatively high counts. In fact, Butte County is one of the small counties, yet it experienced a notably high wildfire number.

Comparing employment trends between Butte and Shasta counties through EDA provides key context for interpreting the DID results. Both Butte and Shasta counties saw steady annual employment levels increase from 2015-2018 (Figure 2), indicating gradual economic growth. After the Camp Fire, which began in November of 2018, these trends diverge: employment levels in Butte County begin to fall, while Shasta County continues to see an increase through 2019. Following the wildfire, employment levels in Butte County fell to below pre-fire levels throughout much of 2019, and appear to recover around October 2019, when the employment level slightly exceeds the pre-fire benchmark of around 83,858 employees (Figure 4). Based only on the observed employment levels within the available dataset, it took about one year for employment to surpass the original pre-fire level. Additional economic indicators and longer-term data would be needed to assess the full economic recovery trajectory. Compared to Butte County, the 2019 economic decline in Shasta County appears less severe, indicating broader regional or statewide factors may have contributed to the decrease in employment, rather than wildfire-specific impacts. This comparison supports the interpretation that the sharper decline observed in Butte County may be associated with the 2018 Camp Fire and its long-term economic consequences.

To estimate the economic impact of the 2018 wildfire, we applied a DID framework comparing employment trends in Butte County (treatment group) and Shasta County (control group) before and after the wildfire. The key assumption of this approach is that employment trends in both counties would have followed similar paths in the absence of the wildfire.

Before 2018, both counties show a similar upward trend in employment, supporting the parallel trends assumption. After the 2018 Camp Fire, employment in Butte County declined considerably, while Shasta County experienced a modest decrease. This divergence suggests that the wildfire had a negative effect on employment in Butte County beyond broader regional economic trends. These findings provide evidence that large wildfire events can lead to sustained economic disruptions at the local level.

Conclusion

This study examined the economic impact of the 2018 Camp Fire on employment in Butte County using EDA and a DID estimation model with Shasta County as the control group. The pre-fire trends appeared relatively similar, supporting the parallel trends assumption. After the wildfire, Butte County's employment declined and stayed below pre-fire levels for much of 2019, with observed recovery in October 2019. The DID results found an additional statistically significant decline of about 3,010 employees in Butte County relative to Shasta County. This suggests the Camp Fire had a substantial short-term negative impact on employment.

However, the analysis is limited because it uses county-level data and only one control county. Future research should include more control regions, longer-term data, and additional economic indicators such as wages, migration, business closures, and industry-level effects.

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